

Problem 1

1. On the Gisette dataset, train decision trees of maximum depth $1, 2, \dots$ up to 12, for a total of 12 decision trees. Use the trained trees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test misclassification errors vs tree depth as two separate curves. Report in a table the minimum test error and the tree depth for which the minimum was attained.

We do this in Python, and obtain a plot of the training and test misclassification errors vs tree depth in figure 1:

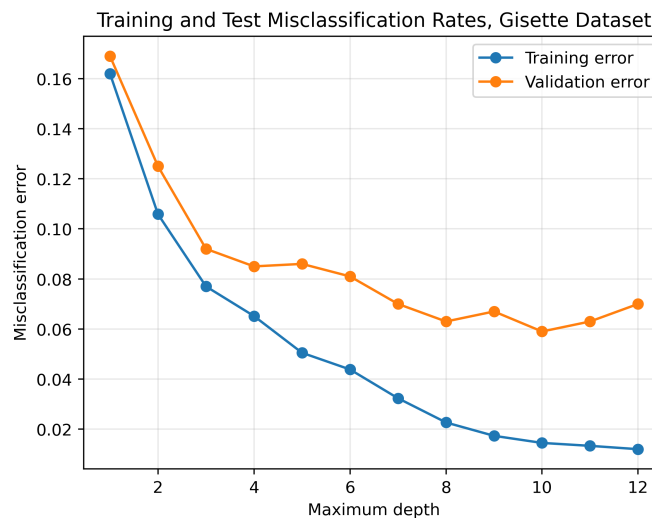


Figure 1: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

The table reporting the minimum test error and the tree depth for which the minimum was attained are displayed in table 1.

Depth	Training Misclassification	Validation Misclassification
10	0.014500	0.059000

Table 1: Lowest misclassification error, Gisette Dataset

In general, the training error decreased as the depth increased. The validation error more or less followed this trajectory, but levelled off for deeper trees.

2. Repeat point 1.) on the satimage, madelon, hill-valley, and covtype datasets.
We can do this also, and obtain the four images shown in figure 2.

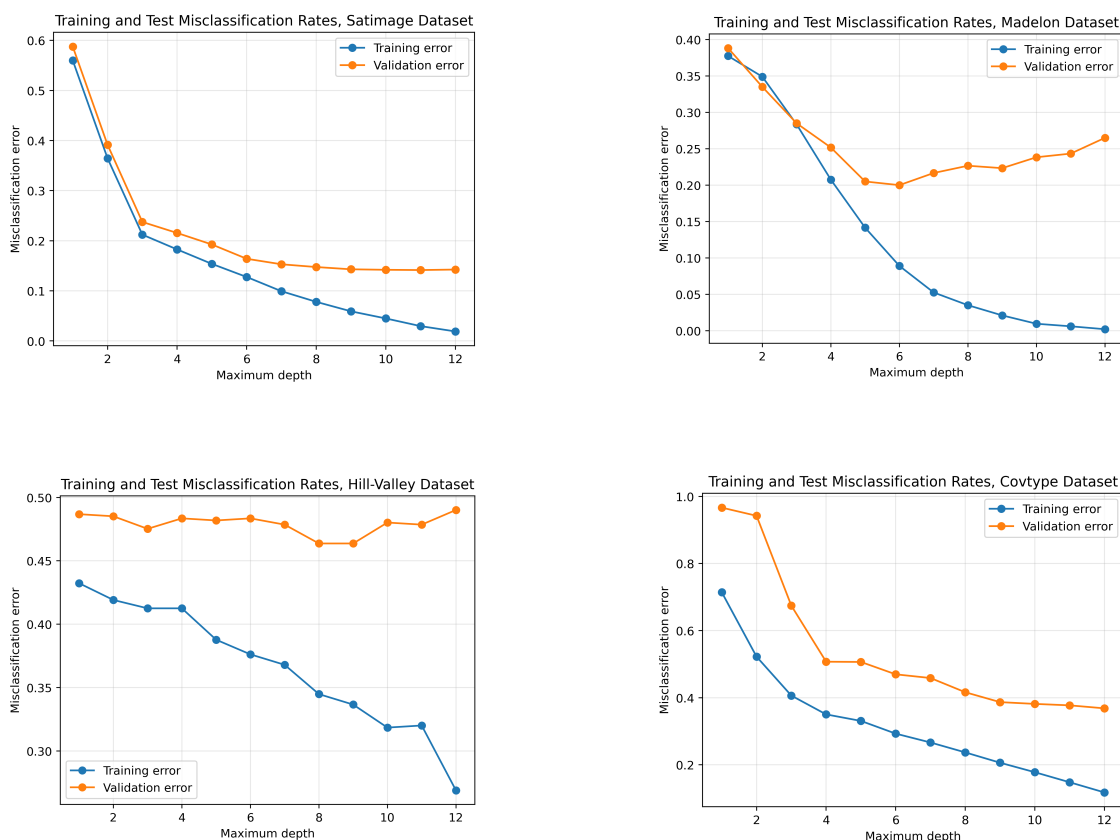


Figure 2: Training and Validation datasets misclassification error rates for four datasets

The table reporting the minimum test error and tree depth for the various datasets is shown in table 2

Dataset	Depth	Training Misclassification	Validation Misclassification
Satimage	11	0.029538	0.141500
Madelon	6	0.089000	0.200000
Hill-Valley	8	0.344884	0.463696
Covtype	12	0.117725	0.368371

Table 2: Lowest misclassification error across datasets

In general, we see the same pattern as with the Gisette dataset. The training error is almost always lower than the validation error.

3. On the Gisette dataset, for each off $k \in \{3, 10, 30, 100, 300\}$ train a random forest with k trees where the split attribute at each node is chosen from a random subset of $71 \approx \sqrt{5000}$ features. Use the trained trees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test errors vs. number of trees k as two separate curves. Report the training and test misclassification errors in a table.

We do this in Python, and obtain the plot of the training and test misclassification errors vs tree depth shown in figure 3:

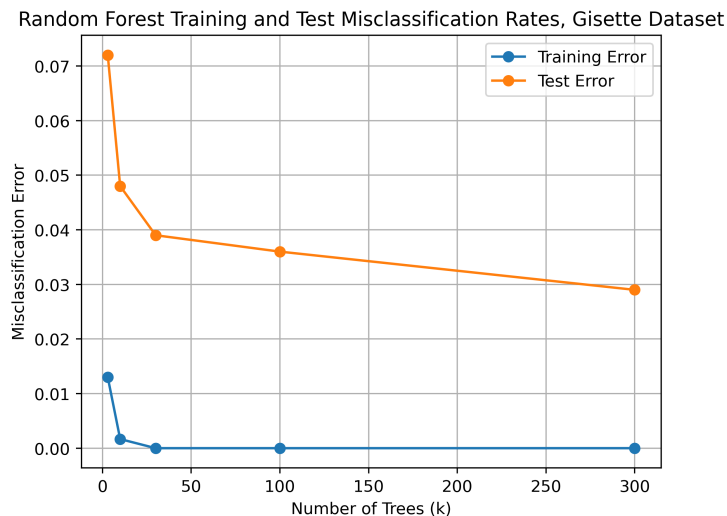


Figure 3: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

The table reporting the training and test errors for various random forests are shown in table 3:

Number of Trees	Training Misclassification Error	Validation Misclassification Error
3	0.013000	0.072000
10	0.001667	0.048000
30	0.000000	0.039000
100	0.000000	0.036000
300	0.000000	0.029000

Table 3: Misclassification errors, Random Forests, Gisette Dataset

Beyond a certain point, the error rates for both the training and the validation datasets do not improve. This also shows the training error being 0 for 100 and 300 trees in the random forest.

4. Repeat point 3.) on the satimage, madelon, and hill-valley datasets.

The images showing the training and test misclassification errors are displayed in figure 4 and we can create a table for this, shown in table 4.

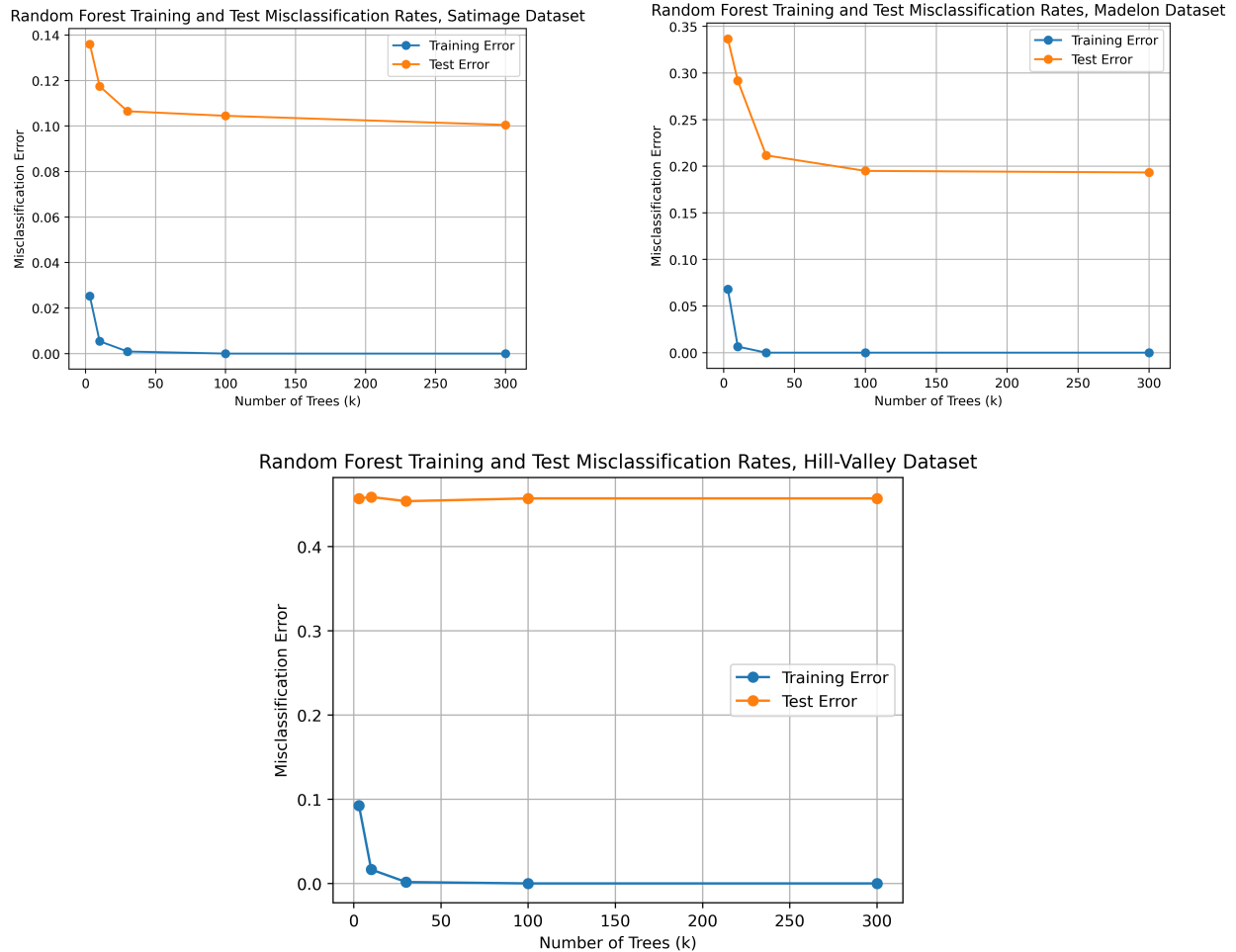


Figure 4: Plots showing training and test misclassification errors versus number of trees.

5. Repeat point 3.) on the Gisette dataset where the split attribute at each node is chosen from a random subset of $12 \approx \log_2(5000)$ features, and where the split attribute at each node is chosen from all 5000 features, plotting all four curves (2 train and 2 test misclassification errors) on the same graph.

Doing this gives us figure 5:

In this plot, we can see that the training error is not meaningfully different when we increase the number of features used. In fact, the validation error is worse for the random forest with 5000 features than for the forest with 12 features.

Dataset	Num. Trees	Training Misclass. Err.	Validation Misclass. Err.
Satimage	3	0.025254	0.136000
Satimage	10	0.005411	0.117500
Satimage	30	0.000902	0.106500
Satimage	100	0.000000	0.104500
Satimage	300	0.000000	0.100500
Madelon	3	0.068000	0.336667
Madelon	10	0.006500	0.291667
Madelon	30	0.000000	0.211667
Madelon	100	0.000000	0.195000
Madelon	300	0.000000	0.193333
Hill-Valley	3	0.092409	0.457096
Hill-Valley	10	0.016502	0.458746
Hill-Valley	30	0.001650	0.453795
Hill-Valley	100	0.000000	0.457096
Hill-Valley	300	0.000000	0.457096

Table 4: Misclassification errors for Random Forests across datasets

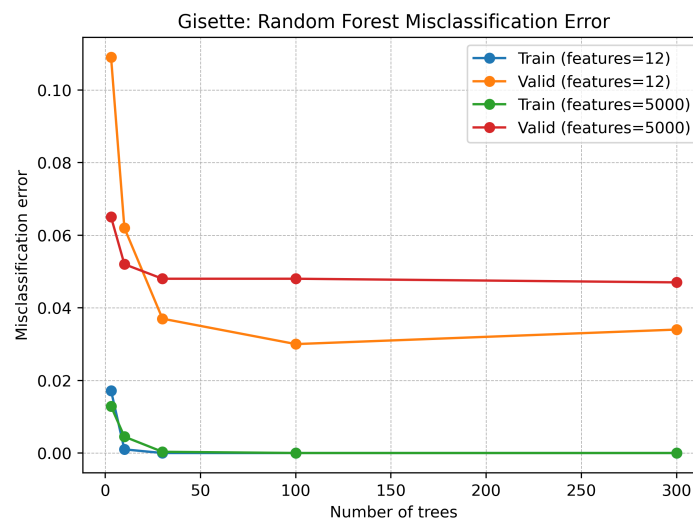


Figure 5: Misclassification Error vs Number of Trees and Number of Features, Training and Validation: Gisette dataset